

Run 3 $H + c (H \rightarrow WW \rightarrow 2\ell 2\nu)$ HWW Meeting

Chirayu Gupta¹, Thomas Van Laer²

Tahys Janssen², Pierre Van Mechelen²

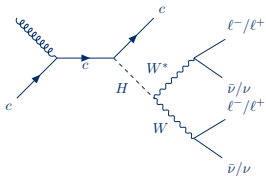
Gerrit Van Onsem¹, Alexandre De Moor¹, Michael Tytgat¹

08/09/2026

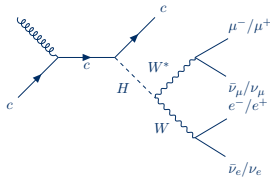
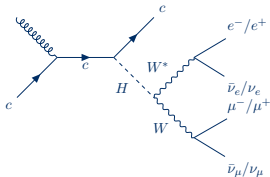
¹Vrije Universiteit Brussel (VUB) ²University of Antwerp

$H + c$ Final State

- Analysis goal: constrain **c Yukawa coupling**
- Look for $H + c$ events in $\mathbf{H(WW \rightarrow 2\ell 2\nu)}$ channel: H couples to c



- Focus on events with **1 electron & 1 muon**



Possibly explore: **dimuon** and **dielectron** final states (possibly insensitive)

Challenges

- Large backgrounds
- Dirty event signature (MET)

Advantages

- High Branching Ratio
- Different dominant uncertainties
- Rich event topology, useful for S vs B MVA training

Run 2 Analysis Strategy (AN-23-102)

Credit: Aachen and Brown group

- Standard WW selection
- Tag a c -jet
- Due to different background compositions, Signal Region (SR) split in 2:
 1. $N_{c\text{-jet}} = 1$
 2. $N_{c\text{-jet}} > 1$

→ Increases sensitivity (8 % for Run 2)
- Define top Control Region (CR) enriched in $t\bar{t}$ events to derive top normalization factor
- Define a high transverse mass CR to build a CR where both $H + c$ and Higgs background contributions are negligible
- Use MVA (Boosted Decision Tree) to optimise signal recognition and separate it from other H production processes
- Modelling of background, main $t\bar{t}$ background in particular, with simultaneous fit in CRs and SRs

Run 2 Results

- Analysis finished (data unblinded) and paper submitted
- Full Run 2 dataset, integrated luminosity: 138 fb^{-1}
- Measured $H + c$ signal strength modifier (μ_{cH}) when:
 - H background fixed to SM (1 POI):

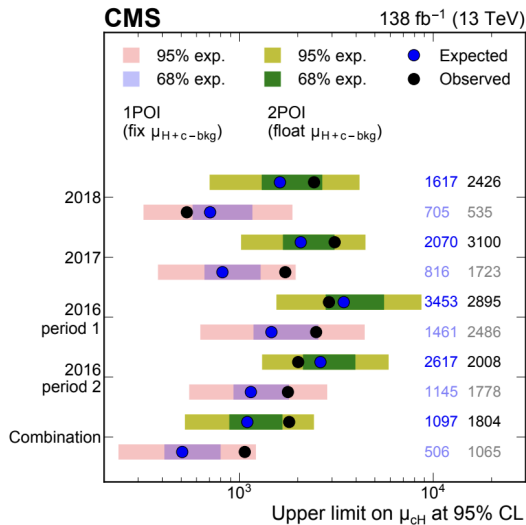
$$390^{+380}_{-270}$$

- H bkg is fit parameter (2 POI):

$$230^{+1000}_{-640}$$

- Observed (expected) upper limit on μ_{cH} at 95 % CL, for 1 POI, is:

$$1065 \text{ (506)}$$



Run 3 Analysis Strategy (WIP)

- Same basic HWW selection as **AN-23-102** (Run 2 analysis)
- $Z \rightarrow \ell\ell$ CR is already validated
- Generate signal MC sample
- Due to different background compositions, Signal Region (SR) split in 2:
 1. $N_{c\text{-jet}} = 1$
 2. $N_{c\text{-jet}} > 1$→ Increases sensitivity (8 % for Run 2)
- Define top CR enriched in $t\bar{t}$ events to derive top normalization factor
- Define a high mass CR to build a CR where both $H + c$ and Higgs background contributions are negligible
- Use MVA (Neural Network) to optimise signal recognition and separate it from other H production processes. Input features inspired by Run 2 analysis
- Modelling of the background, in particular the main $t\bar{t}$ background, with a simultaneous fit in the CRs and SRs using MVA classification

Samples

Currently included: full 2022 and 2023 datasets

Data Samples

- **Muon:** 2022CDEFG, 2023CD from 22Sep2023
(incl. SingleMuon and DoubleMuon for 2022C)
- **EGamma:** 2022CDEFG, 2023CD from 22Sep2023
- **MuonEG:** 2022CDEFG, 2023CD from 22Sep2023

Non-Higgs MC Samples

- **tt**
(powheg, pythia8)
- **Single Top**
(powheg, pythia8)
- **W + Jets**
(amcatnlo FFXFX merging, pythia8)
- **ZZ, WZ, WW**
(amcatnlo FFXFX merging, pythia8)
- **WG**, binned in photon p_T
(amcatnlo FFXFX merging, pythia8)
- **DY + Jets**
(amcatnlo FFXFX merging, pythia8)

Private MC Samples

- **H+c**
(amcatnlo FFXFX merging, pythia8)
- **H+b**
(amcatnlo FFXFX merging, pythia8)

Higgs MC Samples

- **ggH, VBF**
 - **H $\rightarrow$ WW**
(powheg-jhugen752, pythia8)
 - **H $\rightarrow \tau\tau$**
(amcatnlo FFXFX merging, pythia8)
- **ZH**
 - **(qq)ZH**
(powheg-minlo-HZJ-jhugenv752, pythia8)
 - **ggZH**
(powheg-jhugenv752, pythia8)
- **WH**
 - **H $\rightarrow$ WW**
(powheg-minlo-jhugen, pythia8)
 - **H $\rightarrow \tau\tau$**
(powheg-minnlo, pythia8)
- **ttH**
 - **H $\rightarrow$ bb/cc/Z γ**
(powheg, pythia8)
 - **H $\rightarrow \gamma\gamma$**
(amcatnlo FFXFX merging, madspin, pythia8)

Very loose object selections - Base

The basic selection on which the CRs and SRs are based.

Object	Criteria	Selection
Muon	p_T	$> 10 \text{ GeV}$
	$ \eta $	< 2.4
	ID	Tight
	Iso	Tight
Electron	p_T	$> 10 \text{ GeV}$
	$ \eta $	< 2.5
	ID	mvalso_WP80
	$\Delta R(e, \mu)$	> 0.4
Jet	p_T	$> 20 \text{ GeV}$
	$ \eta $	< 2.5
	jetId	$= 6$ (tightLepVeto)
	$\Delta R(J, e/\mu)$	> 0.4
c-jet	is a Jet	ParticleNet b-tagger medium wp
		2022 preEE: $CvB > 0.306$ AND $CvL > 0.160$
	c-tagged	2022 postEE: $CvB > 0.304$ AND $CvL > 0.160$
		2023 preBPix: $CvB > 0.353$ AND $CvL > 0.148$
		2023 postBPix: $CvB > 0.157$ AND $CvL > 0.109$
Lepton Pair	$p_T(\ell_1)$	$> 20 \text{ GeV}$
	$p_T(\ell_2)$	$> 10 \text{ GeV}$
	$q(\ell_1) + q(\ell_2)$	$= 0$
	$m_{\ell\ell}$	$> 12 \text{ GeV}$
	$p_T^{\ell\ell}$	$> 30 \text{ GeV}$

Object selection criteria

Criteria	Selection
Trigger	SingleMu
	SingleEle
	MuEle
MET filters	see below
nPV	> 0
$N(\ell\ell \text{ pair})$	$= 1$
one e , one μ	$\ell\ell$ pair with
	lepton 1 is e and
	lepton 2 is μ or vice versa
p_T^{miss} (Puppi MET)	$> 45 \text{ GeV}$

Event selection criteria

MET Filters (POG Recommendation)

```
goodVertices
globalSuperTightHalo2016Filter
EcalDeadCellTriggerPrimitiveFilter
BadPFMuonFilter
BadPFMuonDzFilter
hfNoisyHitsFilter
eeBadScFilter
ecalBadCalibFilter
```

Triggers

To **avoid duplicate events** from different primary datasets, events are selected as follows:

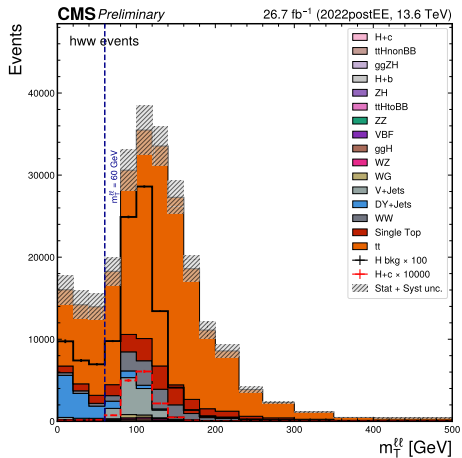
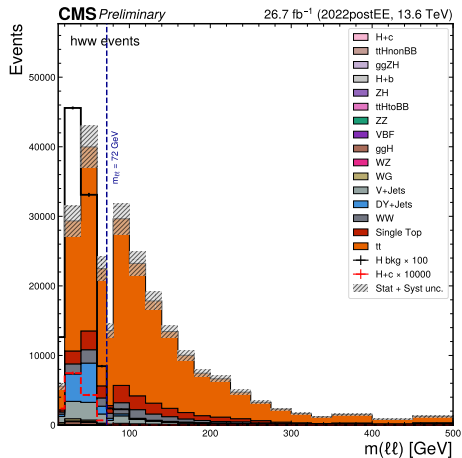
- from **EGamma** if event **passes SingleEle**
- from **Muon** if event **passes SingleMu**, while **failing SingleEle**
- from **MuonEG** if event **passes MuEle** triggers, while **failing SingleEle and SingleMu**

Flag	HLT Paths
SingleMu	IsoMu24
SingleEle	Ele30_WPTight_Gsf
MuEle	Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL Mu12_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ

HLT trigger paths and their corresponding defined flags

Validation of CR Definition

Illustration of choice of cuts used for CR definition



$$m_T^{\ell\ell} = \sqrt{2 \cdot p_T^{\ell\ell} \cdot p_T^{\text{miss}} \cdot (1 - \cos[\Delta\phi(\ell\ell, \text{MET})])}$$

(1)

CRs Selection

- Selections applied on top of Base selections.
- Distinction between the two CRs is shown in red.

Top CR	
Criteria	Selection
$N(\text{c-jet})$	≥ 2
$m_{\ell\ell}$	$> 72 \text{ GeV}$
$m_T^{\ell\ell}$	$\leq 60 \text{ GeV}$
$m_T^{\ell_2}$	$> 30 \text{ GeV}$

High Mass CR	
Criteria	Selection
$N(\text{c-jet})$	≥ 1
$m_{\ell\ell}$	$> 72 \text{ GeV}$
$m_T^{\ell\ell}$	$> 60 \text{ GeV}$
$m_T^{\ell_2}$	$> 30 \text{ GeV}$

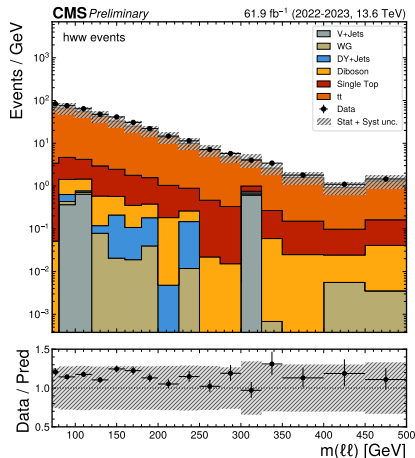
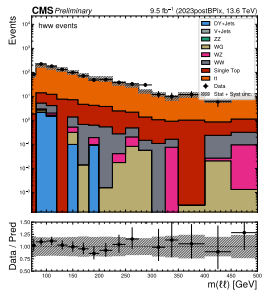
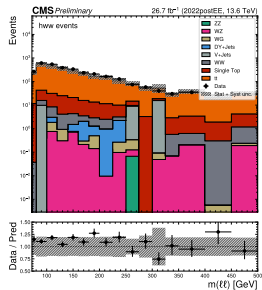
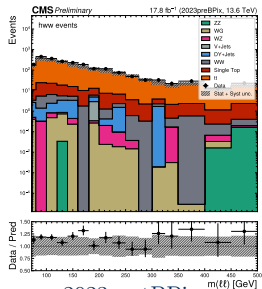
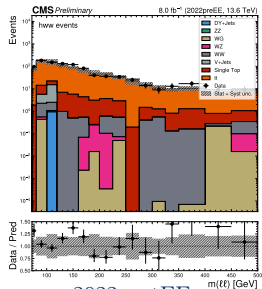
$$m_T^{\ell_2} = \sqrt{2 \cdot p_T^{\ell_2} \cdot p_T^{\text{miss}} \cdot (1 - \cos [\Delta\phi(\ell_2, \text{MET})])} \quad (2)$$

Top Control Region: Dilepton Mass

2022preEE

2023preBPix

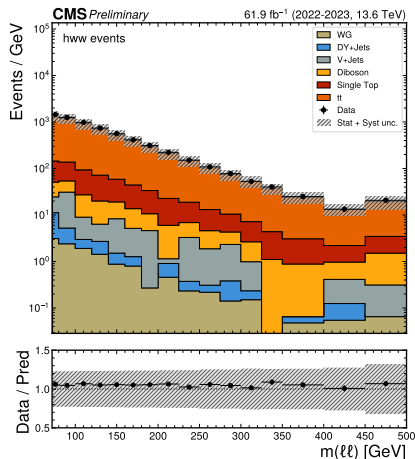
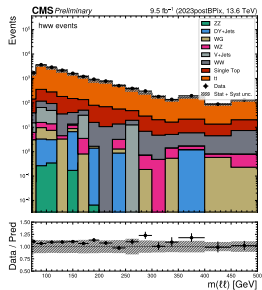
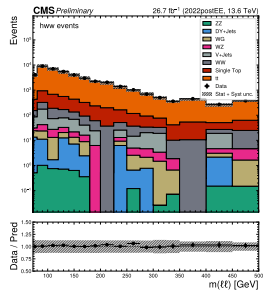
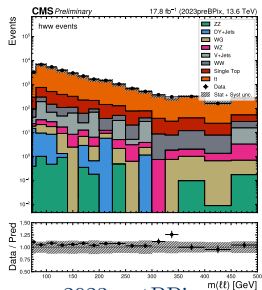
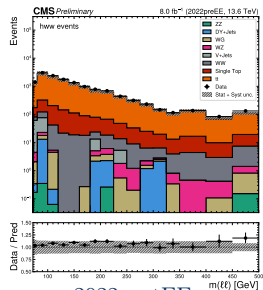
Full 2022-2023



High Mass Control Region: Dilepton Mass

2022preEE 2023preBPix

Full 2022-2023



MVA Training Input Features Based on Run 2

- Run 2: Boosted Decision Tree trained in SR for
 - non-Higgs bkg vs signal
 - Higgs bkg vs signal
- Run 3: Neural Network with several categories:
 - Signal, tt, single top, diboson ...
 - Events not solely in SR: Base selection + ≥ 1 tagged c -jet $\rightarrow$ improves statistics by 15 %

Lepton kinematics	(Transverse) mass	$\Delta\phi$	ΔR
$p_T^{\ell\ell}$	$m_T^{\ell_1}$	$\Delta\phi(\ell\ell + \text{MET}, c)$	$\Delta R(\ell\ell, \ell_1)$
$p_T^{\ell_1}$	$m_T^{\ell_2}$	$\Delta\phi(\ell_1, \text{MET})$	$\Delta R(\ell\ell, \ell_2)$
$p_T^{\ell_2}$	$m_{\ell\ell}$	$\Delta\phi(\ell_2, \text{MET})$	$\Delta R(\ell\ell, c)$
p_T			
Charm tagging	Jet kinematics	Vertex	MET kinematics
c-jet ParticleNet CvL	p_T^c	nSV	p_T^{miss}
c-jet ParticleNet CvB			

α -weighted Feature Importance

Investigate: **which Run 3 features most important** to distinguish Signal from Background.

Based on integrated gradients approach to calculate feature similarity:
checks difference between gradients of a feature for signal vs gradients for a particular background j .

Formula:

$$I_j = \alpha_j \cdot \sum_f \left| \frac{\partial(P_{H+c} - P_j)}{\partial x_f} \right| \text{std}(x_f)$$

where:

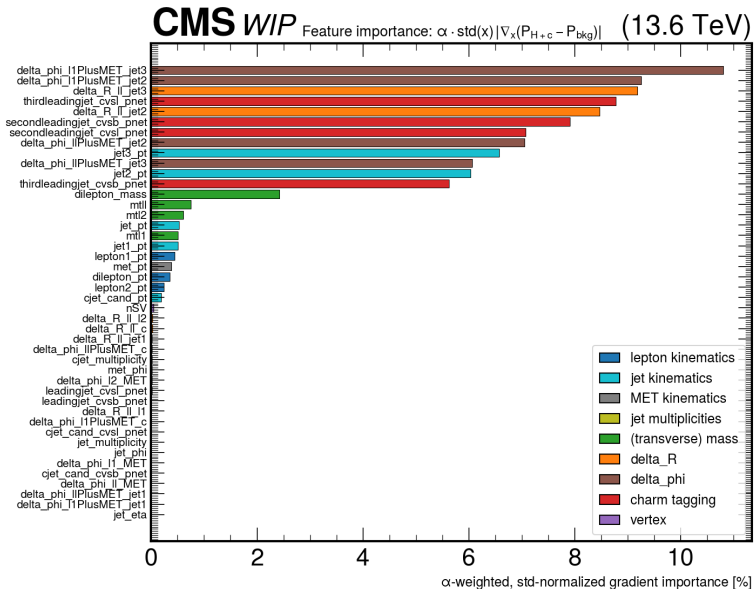
$$\alpha_j = \sigma \left(\frac{\cos(\mathbf{w}_{H+c}, \mathbf{w}_j)}{\tau} \right)$$

See [arXiv:1703.01365](https://arxiv.org/abs/1703.01365).

All Tested Input Features

Lepton kinematics	(Transverse) mass	$\Delta\phi$	ΔR
$p_T^{\ell\ell}$	$m_T^{\ell_1}$	$\Delta\phi(\ell_1, \text{MET})$	$\Delta R(\ell\ell, \ell_1)$
$p_T^{\ell_1}$	$m_T^{\ell_2}$	$\Delta\phi(\ell_2, \text{MET})$	$\Delta R(\ell\ell, \ell_2)$
$p_T^{\ell_2}$	$m_{\ell\ell}$	$\Delta\phi(\ell\ell + \text{MET}, c)$	$\Delta R(\ell\ell, c)$
	$m_T^{\ell\ell}$	$\Delta\phi(\ell\ell, \text{MET})$	$\Delta R(\ell\ell, j_1)$
		$\Delta\phi(\ell_1 + \text{MET}, c)$	$\Delta R(\ell\ell, j_2)$
		$\Delta\phi(\ell_1 + \text{MET}, j_1)$	$\Delta R(\ell\ell, j_3)$
		$\Delta\phi(\ell_1 + \text{MET}, j_2)$	
		$\Delta\phi(\ell_1 + \text{MET}, j_3)$	
		$\Delta\phi(\ell\ell + \text{MET}, j_1)$	
		$\Delta\phi(\ell\ell + \text{MET}, j_2)$	
		$\Delta\phi(\ell\ell + \text{MET}, j_3)$	
Charm tagging	Jet kinematics	Vertex	MET kinematics
c-jet ParticleNet CvL	p_T^c	nSV	p_T^{miss}
c-jet ParticleNet CvB	p_T^{jets}		$\phi(\text{MET})$
jet 1 ParticleNet CvL	$\eta(\text{jets})$		
jet 1 ParticleNet CvB	$\phi(\text{jets})$		
jet 2 ParticleNet CvL	$p_T^{j_1}$		
jet 2 ParticleNet CvB	$p_T^{j_2}$		
jet 3 ParticleNet CvL	$p_T^{j_3}$		
jet 3 ParticleNet CvB			
			Jet multiplicities
			$N(\text{jets})$
			$N(\text{c-jets})$

Feature Ranking



New important features w.r.t.
Run 2 mostly related to 2nd and
3rd leading jets:

- $\Delta\phi(\ell_1 + \text{MET}, \text{jet } 2/3)$
- $\Delta\phi(\ell\ell + \text{MET}, \text{jet } 2/3)$
- $\Delta R(\ell\ell \text{ jet } 2/3)$
- $p_T(\text{jet } 2/3)$
- $CvsL(\text{jet } 2/3)$
- $CvsB(\text{jet } 2/3)$

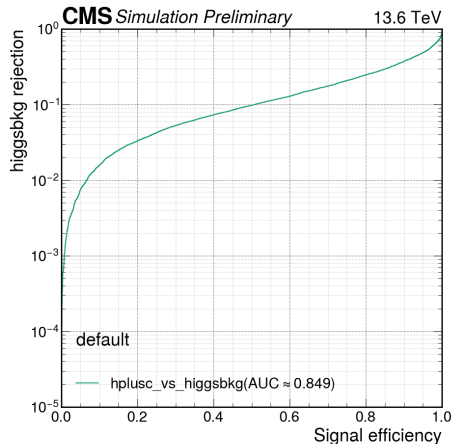
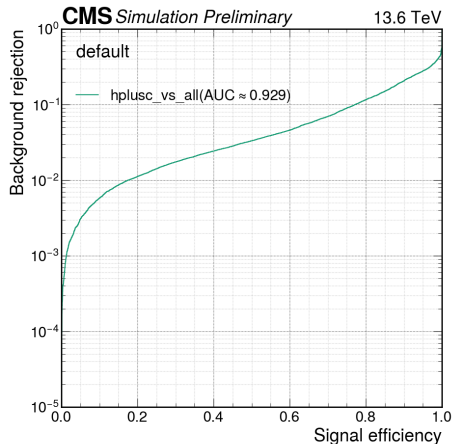
→ Highlights possibility of
removing any features relying on
a c -tagged jet.

MVA Performance

Using basic MVA model: already decent performance.

Signal vs all bkg: $AUC = 0.929$

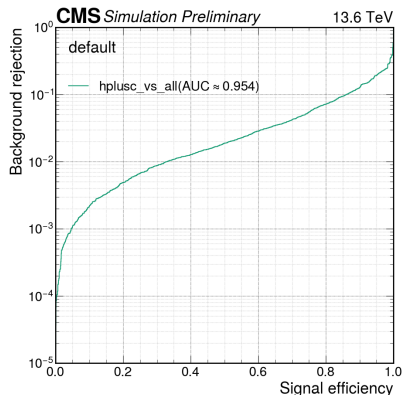
Signal vs H bkg: $AUC = 0.849$



MVA Performance

With a more advanced model, we can improve this bit by bit.
Below: result for only using the 17 features based on Run 2.

Signal vs all bkg: $\text{AUC} = 0.954$



More details on studies for obtaining a better model: Chirayu's talk tomorrow.

Systematic Uncertainties: current inventory

22 shape · 9 lnN · rate_tt free rateParam · autoMCStats (threshold 10) in all 6 channels

Shape — event weight

pileup	pileup profile
ps_isr, ps_fsr	parton shower
scalevar_muR, _muF	renorm./fact. scale
scalevar_muR.muF	correlated variation
lhe_pdf	NNPDF replicas
lhe_alphaS	α_S variation
muon_id, muon_iso	muon efficiency
electron_id	electron efficiency
electron_reco_	reco, p_T -binned
RecoBelow20/20to75/	(<20, 20–75,
RecoAbove75	>75 GeV)
CMS_ctag2d.2022	2D CvL/CvB SF
CMS_negrw_vjets	negative-weight rw.

Shape — object shift

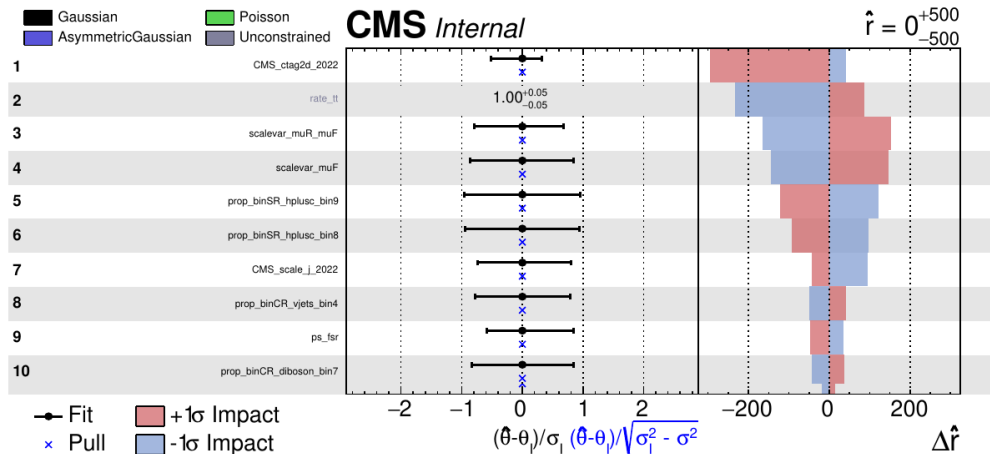
CMS_scale_j.2022	JES
CMS_res_j.2022	JER
CMS_scale_e.2022	electron scale
CMS_res_e.2022	electron res.
CMS_scale_m.2022	muon scale
CMS_res_m.2022	muon res.

lnN

lumi_13p6TeV	1.4%
xsec_st	−1.3/ + 1.7%
xsec_diboson	3.7%
xsec_vjets	2.7%
xsec_higgsbkg	5.0%
flavor_composition_ggH	6.6%
BR_HtoWW, BR_Htautau	1.0%
xsec_hplusc_4FS_5FS	30%

tt̄ carries no xsec term — its normalisation floats freely via rate_tt, pinned by the 94%-pure tt control region.

Nuisance impacts — top 10



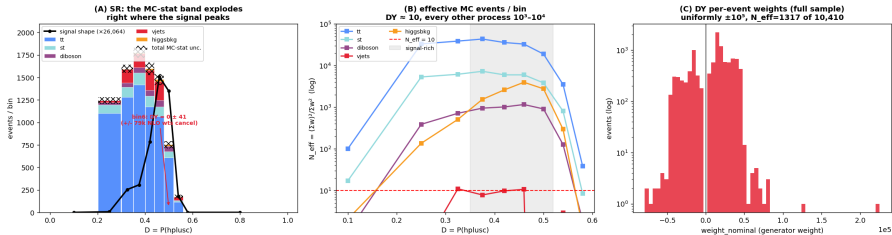
- The header $\hat{r} = 0^{+500}_{-500}$ is a `plotImpacts` rounding artefact — the fit returns $\hat{r} = 0.99996, {}^{+483}_{-513}$, exactly the injected Asimov value

Negative-weight reweighting — the problem

aMC@NLO events carry **signed** weights, so the yield estimator $\hat{N}_B = \sum_{i \in B} w_i$ is a **difference of two large numbers**. The mean is unbiased, but the variance scales with $\sum |w_i|^2$ — the *unsigned* statistics — so the effective sample size $N_{eff} = (\sum w)^2 / \sum w^2$ collapses.

16.4% of our V+jets events carried a negative weight.

autoMCStats blow-up = DY (vjets) is starved in the H+c signal region



(A) the MC-stat band explodes where the signal peaks — one bin was $DY = 0 \pm 41$ from $\pm 79k$ weights cancelling · (B) N_{eff} per bin: V+jets ≈ 10 , every other process 10^3 - 10^4

Negative-weight reweighting — the method

Write the signed generator density as a difference of two *positive* densities, $\text{PDF} = a \text{PDF}_+ - b \text{PDF}_-$ with $a - b = 1$. Defining $P_+(\vec{x}) = a \text{PDF}_+ / (a \text{PDF}_+ + b \text{PDF}_-)$:

$$\text{PDF}(\vec{x}) = \underbrace{(2P_+(\vec{x}) - 1)}_{g(\vec{x})} \cdot \underbrace{(a \text{PDF}_+ + b \text{PDF}_-)}_{\text{unsigned density}}$$

A classifier learns P_+ from generator-level kinematics; each event is then reweighted by $g(\vec{x})$, so the estimator $\sum |w| g$ samples the **unsigned** density instead of differencing large signed sums.

	V+jets (SR)
N_{eff}	280 $\rightarrow$ 1170
statistical error	5.98% $\rightarrow$ 2.92%
rate	1508 $\rightarrow$ 1523

- Training-region closure **0.994** on 9.8M events
- Renormalised **per dataset** — yield restored exactly, variance reduction kept
- Retrain pending on the jet-binned W+jets samples

The rate barely moves while the statistical error halves — the signature of a statistics gain, not a normalisation change.

Method follows **Palmer & Kronheim**, arXiv:2510.16217.

Comparison with the Run 2 analysis

	Expected UL
AN-23-102 scaled to 26.7 fb^{-1}	1144
this analysis	1034
our statistical only	641

- **Better than** the published analysis scaled to our luminosity — a **9.6% improvement** on **5× less data**

Conclusion

- 2022 and 2023 are both included now
- $Z \rightarrow \ell\ell$ control region has been validated (in common with $H \rightarrow ZZ$ analysis)
- MC signal and $H + b$ background samples generated. Central Production underway
- Investigating best input features and architecture for MVA
- Preliminary estimates show that Run 3 analysis already promises to be competitive with Run 2, but we need to stress that this is still very much WIP

Next steps:

- Converge on lepton working points used in the main HWW analysis
- Optimize MVA training using calibrated PNet features
- Study systematic uncertainty when using 3 leading jets instead of 1 tagged c jet
- Extend to 2024 data (Run 3, NanoAODv15) once all MC is available

Backup

2022 Experimental Data Samples

- Below, experimental data samples used from 2022preEE and 2022postEE campaigns

Campaign	DAS Path	Luminosity [1/fb]
2022preEE	Muon/Run2022C-22Sep2023-v1/NANOAOD	5.0104
	DoubleMuon/Run2022C-22Sep2023-v1/NANOAOD	
	SingleMuon/Run2022C-22Sep2023-v1/NANOAOD	
	EGamma/Run2022C-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022C-22Sep2023-v1/NANOAOD	2.9700
	Muon/Run2022D-22Sep2023-v1/NANOAOD	
	EGamma/Run2022D-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022D-22Sep2023-v1/NANOAOD	
2022postEE	Muon/Run2022E-22Sep2023-v1/NANOAOD	5.8070
	EGamma/Run2022E-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022E-22Sep2023-v1/NANOAOD	
	Muon/Run2022F-22Sep2023-v2/NANOAOD	17.7819
	EGamma/Run2022F-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022F-22Sep2023-v1/NANOAOD	
	Muon/Run2022G-22Sep2023-v1/NANOAOD	3.0828
	EGamma/Run2022G-22Sep2023-v2/NANOAOD	
	MuonEG/Run2022G-22Sep2023-v1/NANOAOD	

2023preBPix Experimental Data Samples

- Below, experimental data samples used from 2023preBPix campaign

Campaign	DAS Path	Luminosity [1/fb]
2023preBPix	MuonEG/Run2023C-22Sep2023_v1-v1/NANOAOD	17.7940
	MuonEG/Run2023C-22Sep2023_v2-v1/NANOAOD	
	MuonEG/Run2023C-22Sep2023_v3-v1/NANOAOD	
	MuonEG/Run2023C-22Sep2023_v4-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v1-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v2-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v3-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v4-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v1-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v2-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v3-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v4-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v1-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v2-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v3-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v4-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v1-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v2-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v3-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v4-v2/NANOAOD	

2023postBPix Experimental Data Samples

- Below, experimental data samples used from 2023postBPix campaign

Campaign	DAS Path	Luminosity [1/fb]
2023postBPix	MuonEG/Run2023D-22Sep2023_v1-v1/NANOAOD	9.4500
	MuonEG/Run2023D-22Sep2023_v2-v1/NANOAOD	
	EGamma0/Run2023D-22Sep2023_v1-v1/NANOAOD	
	EGamma0/Run2023D-22Sep2023_v2-v1/NANOAOD	
	EGamma1/Run2023D-22Sep2023_v1-v1/NANOAOD	
	EGamma1/Run2023D-22Sep2023_v2-v1/NANOAOD	
	Muon0/Run2023D-22Sep2023_v1-v1/NANOAOD	
	Muon0/Run2023D-22Sep2023_v2-v1/NANOAOD	
	Muon1/Run2023D-22Sep2023_v1-v1/NANOAOD	
	Muon1/Run2023D-22Sep2023_v2-v1/NANOAOD	

Background MC Samples 2022 and 2023

Process	DAS Path	$\sigma \times \text{BR}$ [pb]	Source
tt			
$tt \rightarrow 2W + 2b \rightarrow 2\ell 2\nu + 2b$	TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	98.578	Twiki for NNLO σ , arxiv:2201.07861 for BR
$tt \rightarrow 2W + 2b \rightarrow (\ell\nu + 2q) + 2b$	TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8	406.263	Twiki for NNLO σ , arxiv:2201.07861 for BR
$tt \rightarrow 2W + 2b \rightarrow 4q + 2b$	TTto4Q_TuneCP5_13p6TeV_powheg-pythia8	418.574	Twiki for NNLO σ , arxiv:2201.07861 for BR
Single Top			
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow 2q + b + 2q$	TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8	19.918	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow \ell\nu + b + 2q$	TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8	19.332	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow \ell\nu + b + \ell\nu$	TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	4.691	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow 2q + \bar{b} + 2q$	TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8	19.918	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow \ell\nu + \bar{b} + 2q$	TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8	19.332	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow \ell\nu + \bar{b} + \ell\nu$	TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	4.691	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + q \rightarrow (W^- + \bar{b}) + q \rightarrow 2q + \bar{b} + q$	TbarQtto2Q-t-channel_TuneCP5_13p6TeV_powheg-pythia8	58.703	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + q \rightarrow (W^- + \bar{b}) + q \rightarrow \ell\nu + \bar{b} + q$	TbarQttoLNu-t-channel_TuneCP5_13p6TeV_powheg-pythia8	28.488	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + \bar{q} \rightarrow (W^+ + b) + \bar{q} \rightarrow 2q + b + \bar{q}$	TQbarto2Q-t-channel_TuneCP5_13p6TeV_powheg-pythia8	97.614	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + \bar{q} \rightarrow (W^+ + b) + \bar{q} \rightarrow \ell\nu + b + \bar{q}$	TQbartoLNu-t-channel_TuneCP5_13p6TeV_powheg-pythia8	47.372	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + b \rightarrow (W^- + \bar{b}) + b \rightarrow \ell\nu + \bar{b} + b$	TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8	1.481	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + \bar{b} \rightarrow (W^+ + b) + \bar{b} \rightarrow \ell\nu + b + \bar{b}$	TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8	2.367	Twiki for NNLO σ , arxiv:2201.07861 for BR
V + Jets			
$W + 0, 1, 2 \text{ jets} \rightarrow \ell\nu + \text{jets}$	WtoLNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	67710.000	XSDB
Diboson			
$ZZ \rightarrow \dots$	ZZ_TuneCP5_13p6TeV_powheg-pythia8	16.70	CMS AN-24-078
$WZ \rightarrow \dots$	WZ_TuneCP5_13p6TeV_powheg-pythia8	54.30	CMS AN-24-078
$WW \rightarrow \dots$	WW_TuneCP5_13p6TeV_powheg-pythia8	122.30	CMS AN-24-078

Background MC samples and cross sections values for both 2022 and 2023

Background MC Samples 2022 and 2023

Process	DAS Path	$\sigma \times \text{BR}$ [pb]	Source
WG (Diboson)			
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUG-1Jets_PTG-10to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	662.2	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUG-1Jets_PTG-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	2.221	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUG-1Jets_PTG-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	0.2908	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUG-1Jets_PTG-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	0.02231	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUG-1Jets_PTG-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	0.004907	XSDB
DY + Jets			
$q\bar{q} \rightarrow Z/\gamma^* + 0, 1, 2 \text{ jets} \rightarrow 2\ell + \text{jets}$	DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	6688.0	XSDB
$q\bar{q} \rightarrow Z/\gamma^* + 0, 1, 2 \text{ jets} \rightarrow 2\ell + \text{jets}$	DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	20950.0	XSDB
Higgs Production			
$g\bar{g} \rightarrow H \rightarrow WW \rightarrow 2\ell 2\nu$	GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8	1.0825	YR4 SM for σ , arxiv:2201.07861 for BR
$q\bar{q} + VV \rightarrow q\bar{q} + H \rightarrow q\bar{q} + WW \rightarrow q\bar{q} + 2\ell 2\nu$	VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8	0.09305	YR4 SM for σ , arxiv:2201.07861 for BR
$q\bar{q} \rightarrow Z + H \rightarrow \dots + WW \rightarrow \dots + 2\ell 2\nu$	ZH_ZtoAll_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-HZJ-jhugenv752-pythia8	0.021559	YR4 SM for σ , arxiv:2201.07861 for BR
$g\bar{g} \rightarrow Z + H \rightarrow \dots + WW \rightarrow \dots + 2\ell 2\nu$	GluGluZH_ZtoAll_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugenv752-pythia8	0.003094	YR4 SM for σ , arxiv:2201.07861 for BR
$g\bar{g} \rightarrow H \rightarrow 2\tau$	GluGluHto2Tau_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	2.976691	YR4 SM for σ , arxiv:2201.07861 for BR
$q\bar{q} + VV \rightarrow q\bar{q} + H \rightarrow 2\tau$	VBFHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.255873	YR4 SM for σ , arxiv:2201.07861 for BR
$W^+ + H \rightarrow \ell\nu + WW \rightarrow \ell\nu + 2\ell 2\nu$	WplusH_WtoLNU_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-jhugen-pythia8	0.006625	YR4 SM for σ , arxiv:2201.07861 for BR
$W^- + H \rightarrow q\bar{q} + WW \rightarrow q\bar{q} + 2\ell 2\nu$	WminusH_Wto2Q_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-jhugen-pythia8	0.008719	YR4 SM for σ , arxiv:2201.07861 for BR
$W^- + H \rightarrow \dots + 2\tau$	WminusHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8	0.035615	YR4 SM for σ , arxiv:2201.07861 for BR
$W^+ + H \rightarrow \dots + 2\tau$	WplusHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8	0.055763	YR4 SM for σ , arxiv:2201.07861 for BR
$t\bar{t} + H \rightarrow \dots + 2c$	ttHto2C_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.016509	YR4 SM for σ , arxiv:2201.07861 for BR
$t\bar{t} + H \rightarrow \dots + 2\gamma$	ttHtoGG_M-125_TuneCP5_13p6TeV_amcatnloFXFX-madspin-pythia8	0.001296	YR4 SM for σ , arxiv:2201.07861 for BR
$t\bar{t} + H \rightarrow \dots + (Z + \gamma) \rightarrow \dots + \dots + \gamma$	ttHtoZG_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.000875	YR4 SM for σ , arxiv:2201.07861 for BR
$t\bar{t} + H \rightarrow \dots + 2b$	ttHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.332585	YR4 SM for σ , arxiv:2201.07861 for BR
$H + b \rightarrow WW + b \rightarrow 2\ell 2\nu + b$	private sample	0.002214	
$H + c \rightarrow WW + b \rightarrow 2\ell 2\nu + b$	private sample	0.016972	

Background MC samples (+ private signal MC sample) and cross sections values for both 2022 and 2023

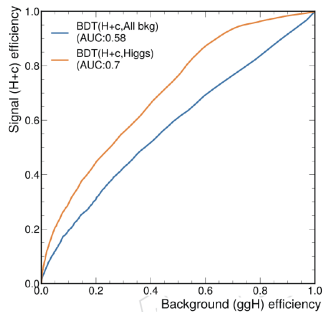
Missing Higgs MC Samples 2022 and 2023

Process	Sample(s)	Status / Remarks
$H + c$	HPlusCharm	In central production
$H + b$	HPlusBottom	In central production
$W + H \rightarrow \ell\nu + WW \rightarrow \ell\nu + 2\ell 2\nu$	WplusH_WtoLNu_Hto2Wto2L2Nu, WminusH_WtoLNu_Hto2Wto2L2Nu	WplusH available for 2022postEE, not for other years
$W + H \rightarrow qq + WW \rightarrow qq + 2\ell 2\nu$	WplusH_Wto2Q_Hto2Wto2L2Nu, WminusH_Wto2Q_Hto2Wto2L2Nu	WminusH available for 2022postEE, not for other years
$tt + H \rightarrow \dots + \text{non-bb}$	ttHToNonbb	Other non-bb samples, besides $2c$, 2γ , $Z\gamma$ needed?
$Z + H \rightarrow \dots + 2\tau$	ZHToTauTau / ZHTo2Tau	Now available, but not yet used
$tt + H \rightarrow \dots + 2\tau$	ttHToTauTau / ttHTo2Tau	Not available for all Run 3 years

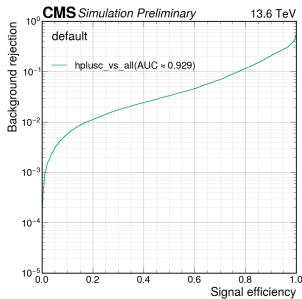
Missing MC samples for both 2022 and 2023

Run 2 vs Run 3 (v11) MVA

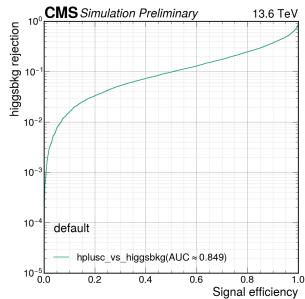
Run 2



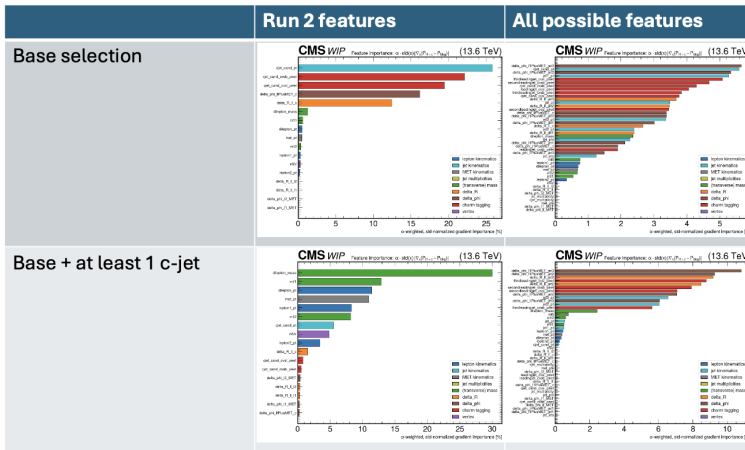
Run 3 Higgs vs all



Run 3 Higgs vs Higgs bkg



Input feature analysis for 2022preEE - normalized



The selection of the c-jet is done with the medium working point from ParticleNet. Data is for 2022preEE.

Input feature analysis for 2022preEE

AUC overview

MVA model v6

Hplusc_vs_all	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	1.01

Hplusc_vs_diboson	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	0.993

Hplusc_vs_higgsgk	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	1.02

Hplusc_vs_st	All possible / Run 2
Base selection	1.00
Base + at least 1 c-jet	0.997

Hplusc_vs_tt	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	1.01

Hplusc_vs_vjets	All possible / Run 2
Base selection	1.00
Base + at least 1 c-jet	1.00

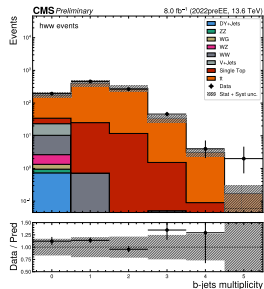
For some MVA categories the one selection is better, but for others, the other one is better. Generally the difference becomes larger in the worst performing categories and there the selection with the cjet cut does better.

Input feature analysis for 2022preEE

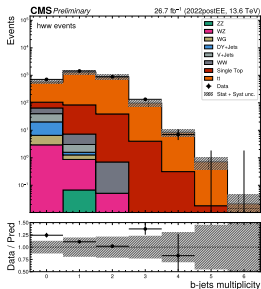
	Base selection	Base + at least 1 c-jet	Improvement by staying with Base
S/B	0.0043	0.0022	1.95
Significance S/sqrt(B)	7.29	2.48	2.94

Top Control Region: `bjet_multiplicity`

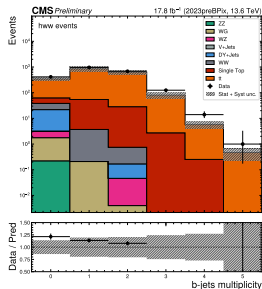
2022preEE



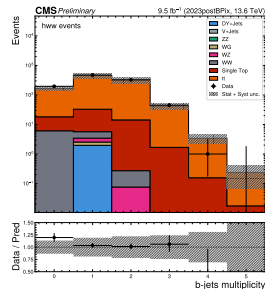
2022postEE



2023preBPix

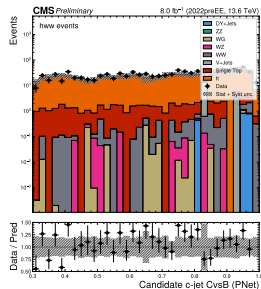


2023postBPix

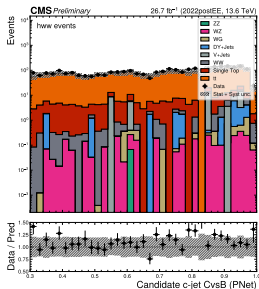


Top Control Region: cjet_cand_cvsb_pnet

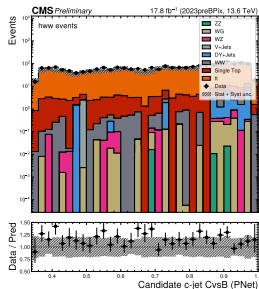
2022preEE



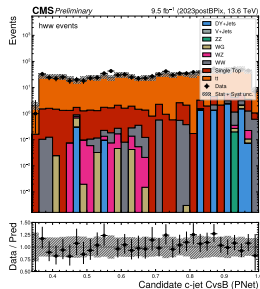
2022postEE



2023preBPix

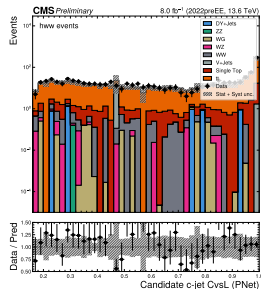


2023postBPix

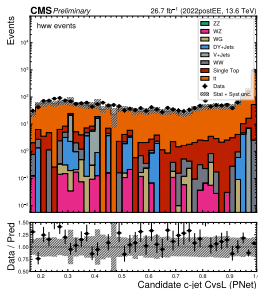


Top Control Region: cjet_cand_cvsl_pnet

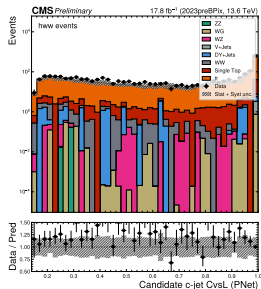
2022preEE



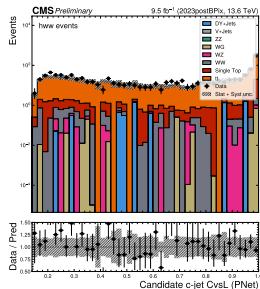
2022postEE



2023preBPix

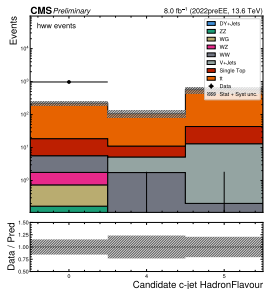


2023postBPix

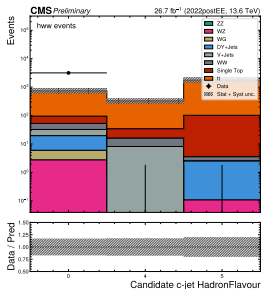


Top Control Region: cjet_cand_flavour

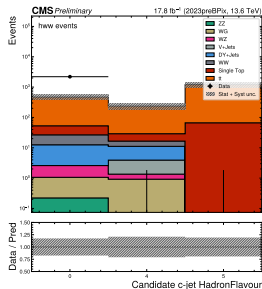
2022preEE



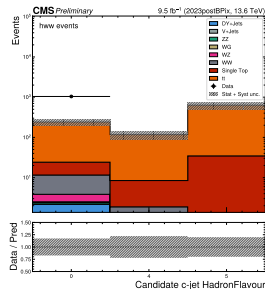
2022postEE



2023preBPix

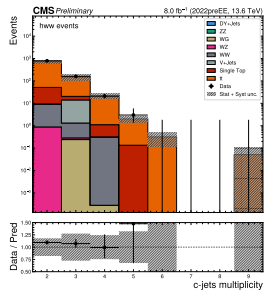


2023postBPix

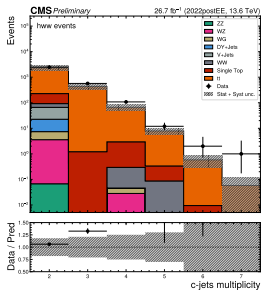


Top Control Region: `cjet_multiplicity`

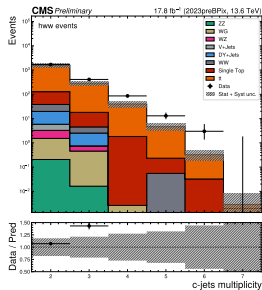
2022preEE



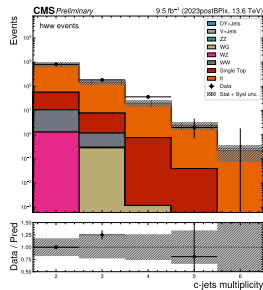
2022postEE



2023preBPix

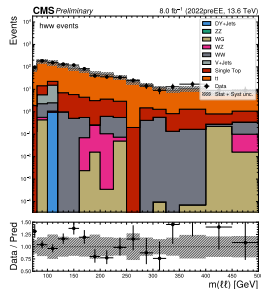


2023postBPix

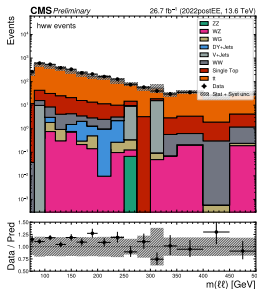


Top Control Region: dilepton mass

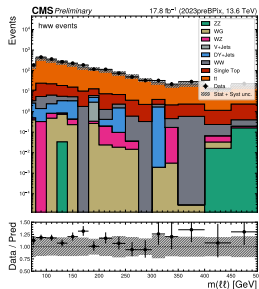
2022preEE



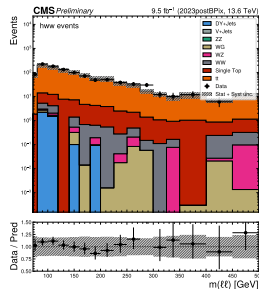
2022postEE



2023preBPix

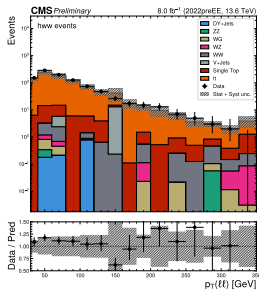


2023postBPix

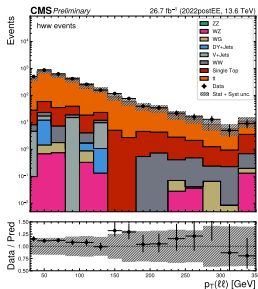


Top Control Region: dilepton_pt

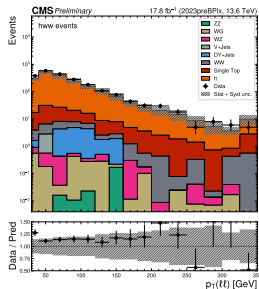
2022preEE



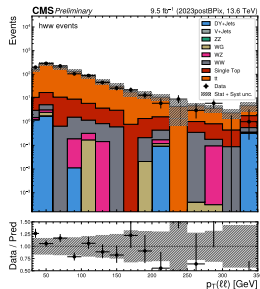
2022postEE



2023preBPix

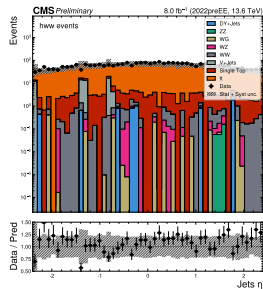


2023postBPix

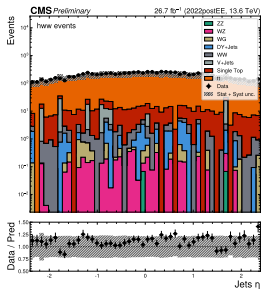


Top Control Region: jet_eta

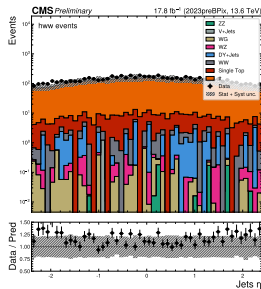
2022preEE



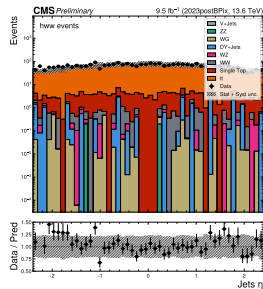
2022postEE



2023preBPix

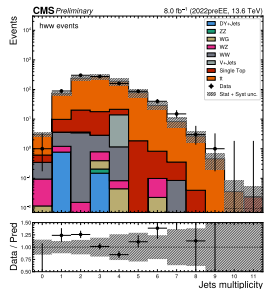


2023postBPix

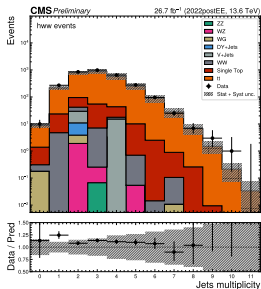


Top Control Region: jet multiplicity

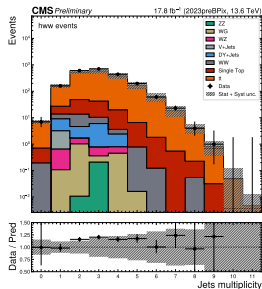
2022preEE



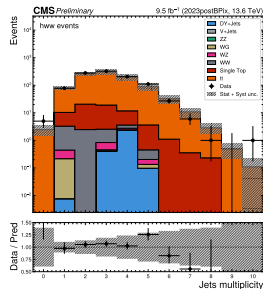
2022postEE



2023preBPix

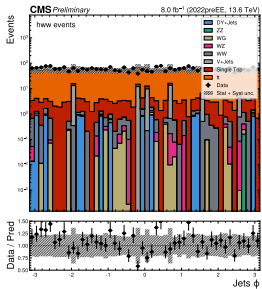


2023postBPix

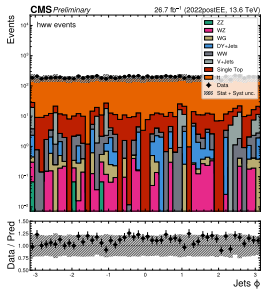


Top Control Region: jet_phi

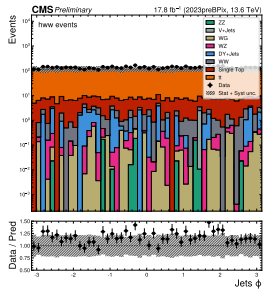
2022preEE



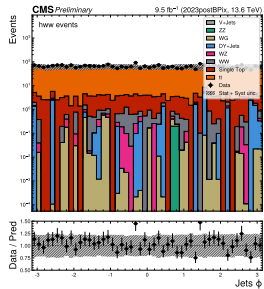
2022postEE



2023preBPix

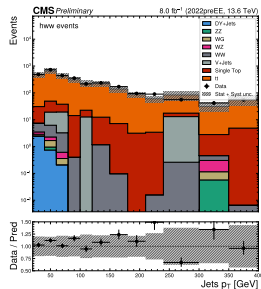


2023postBPix

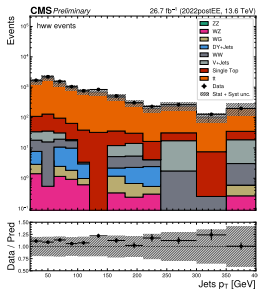


Top Control Region: jet_pt

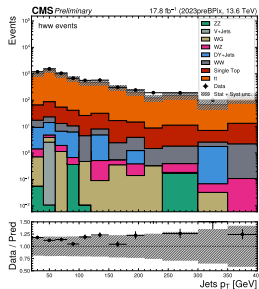
2022preEE



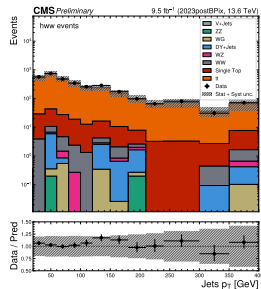
2022postEE



2023preBPix

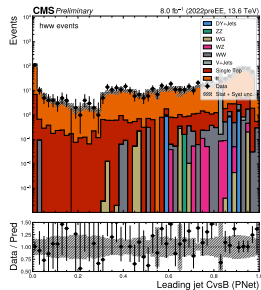


2023postBPix

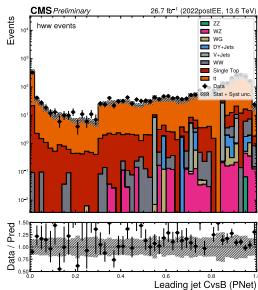


Top Control Region: leadingjet_cvsb_pnet

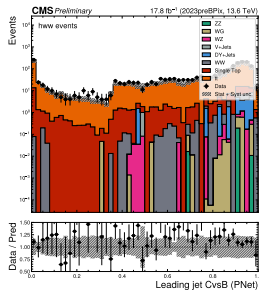
2022preEE



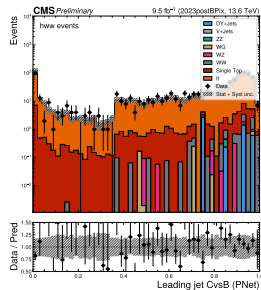
2022postEE



2023preBPix

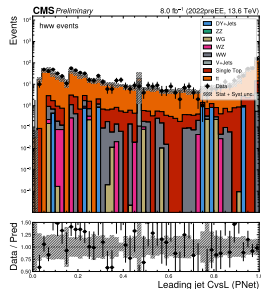


2023postBPix

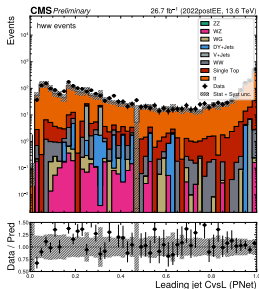


Top Control Region: leadingjet_cvsl_pnet

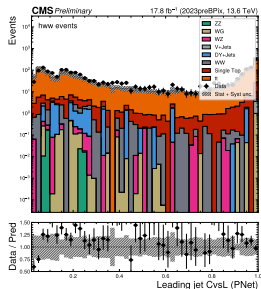
2022preEE



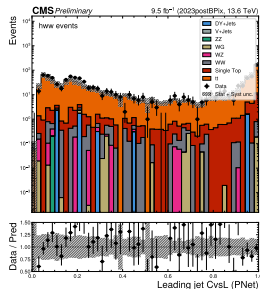
2022postEE



2023preBPix

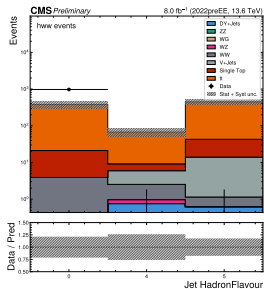


2023postBPix

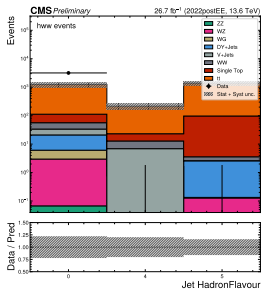


Top Control Region: leadingjet_flavour

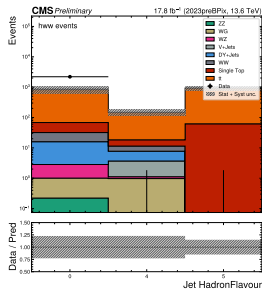
2022preEE



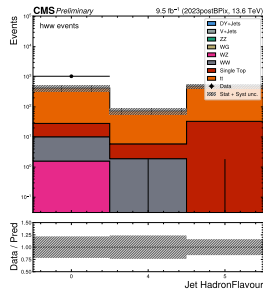
2022postEE



2023preBPix

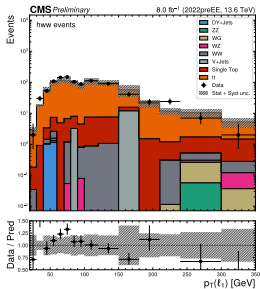


2023postBPix

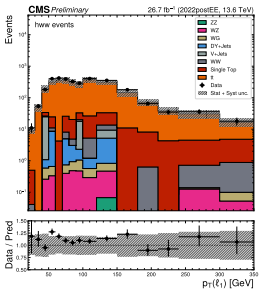


Top Control Region: $1e_{\text{pton1_pt}}$

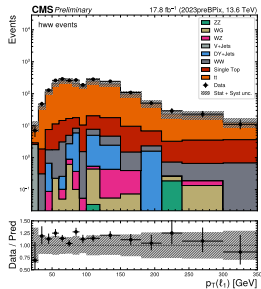
2022preEE



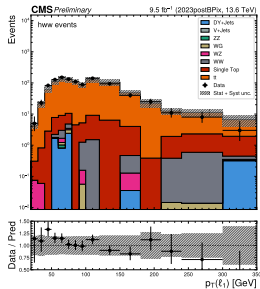
2022postEE



2023preBPix

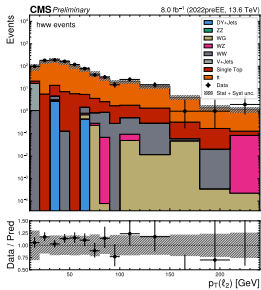


2023postBPix

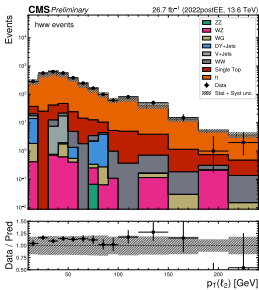


Top Control Region: $1e\mu\nu 2_{pt}$

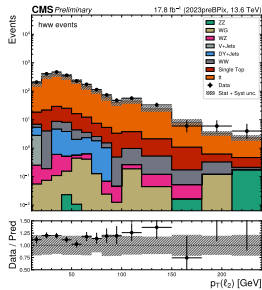
2022preEE



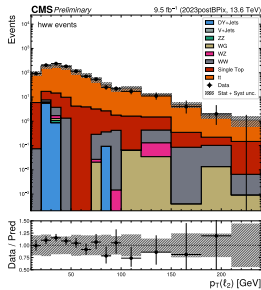
2022postEE



2023preBPix

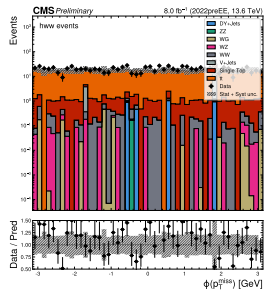


2023postBPix

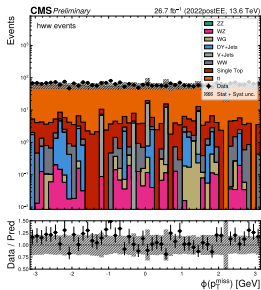


Top Control Region: met_phi

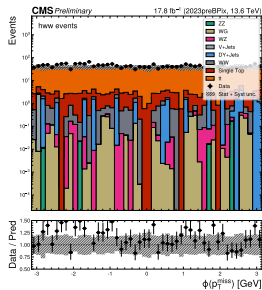
2022preEE



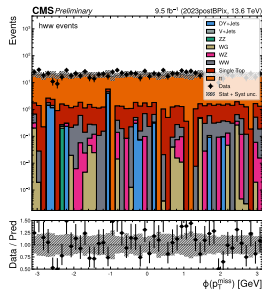
2022postEE



2023preBPix

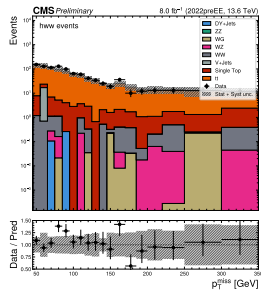


2023postBPix

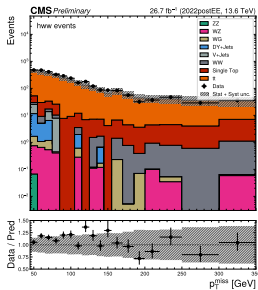


Top Control Region: met_pt

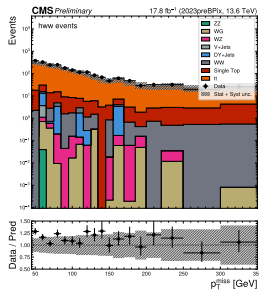
2022preEE



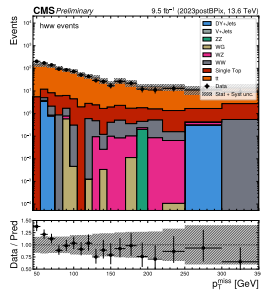
2022postEE



2023preBPix

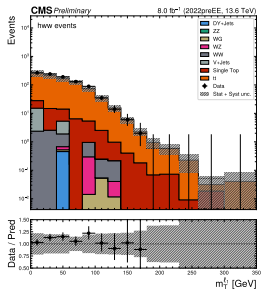


2023postBPix

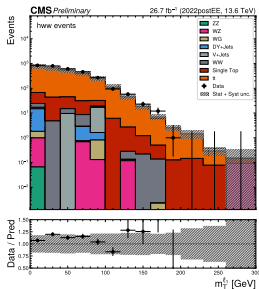


Top Control Region: mt11

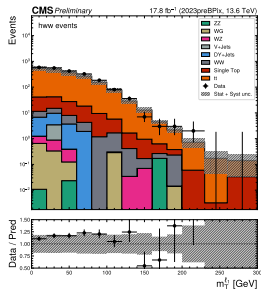
2022preEE



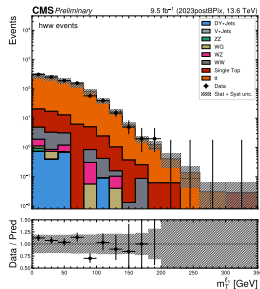
2022postEE



2023preBPix

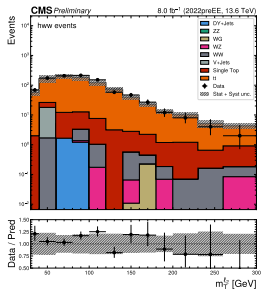


2023postBPix

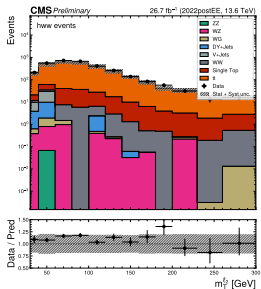


Top Control Region: mt12

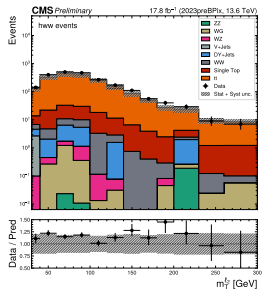
2022preEE



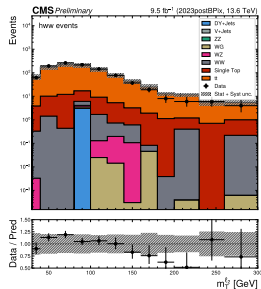
2022postEE



2023preBPix

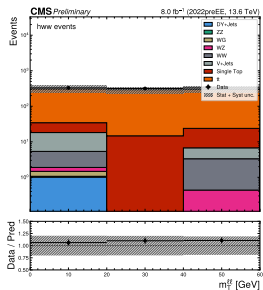


2023postBPix

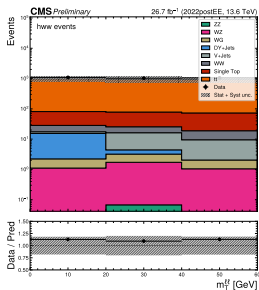


Top Control Region: mt11

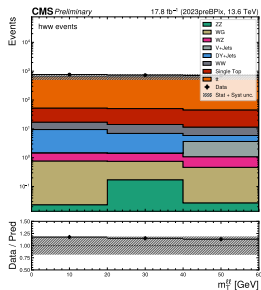
2022preEE



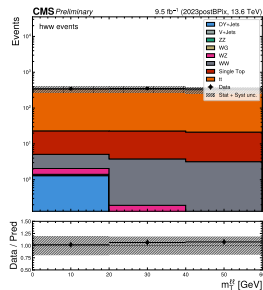
2022postEE



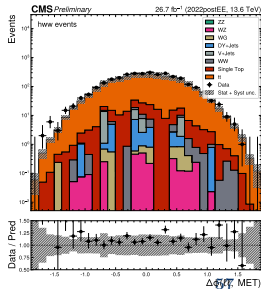
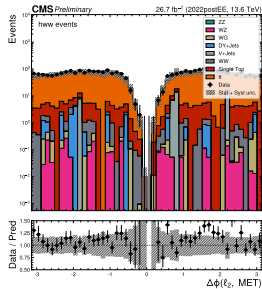
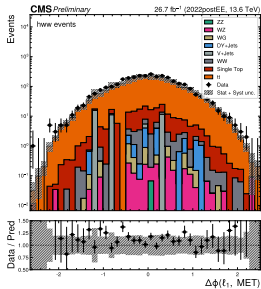
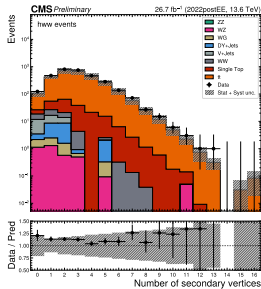
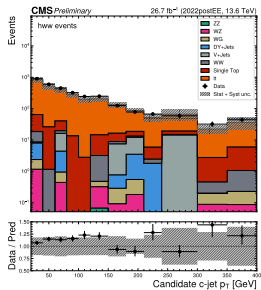
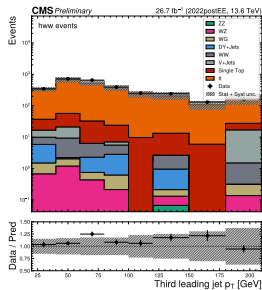
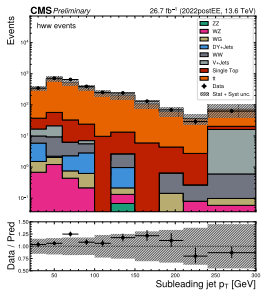
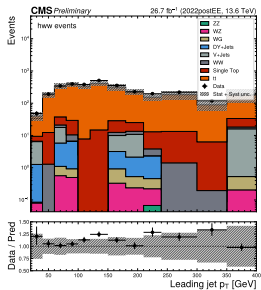
2023preBPix



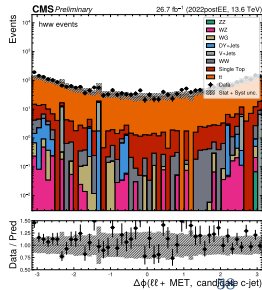
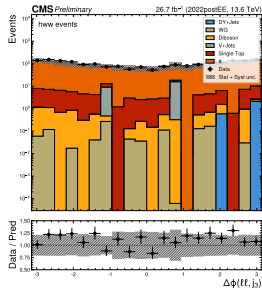
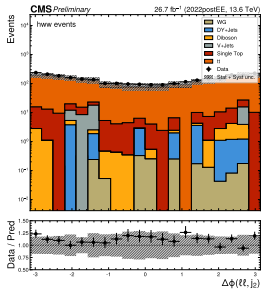
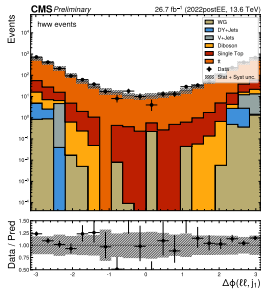
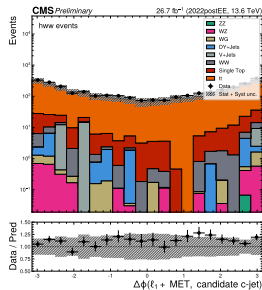
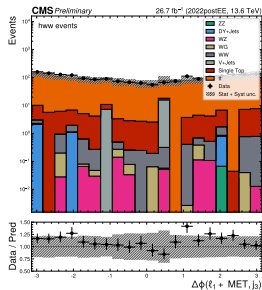
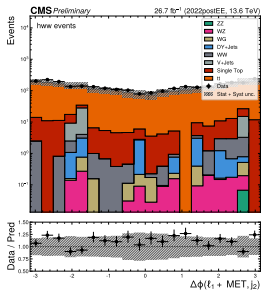
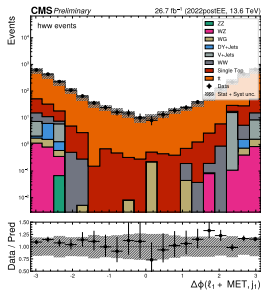
2023postBPix



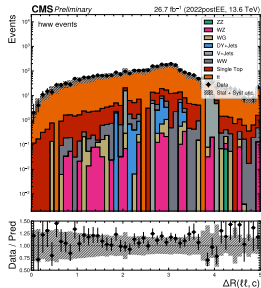
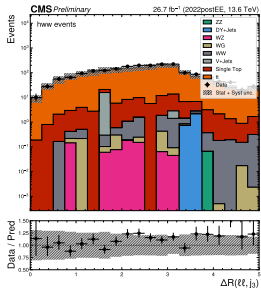
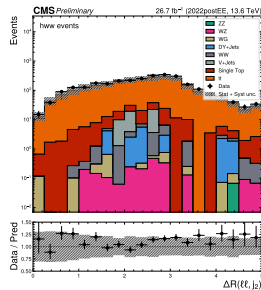
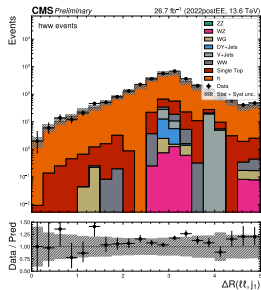
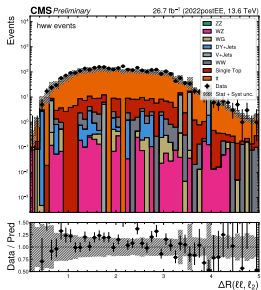
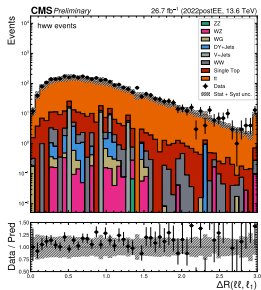
Top CR 2022postEE - extra MVA variables



Top CR 2022postEE - extra MVA variables

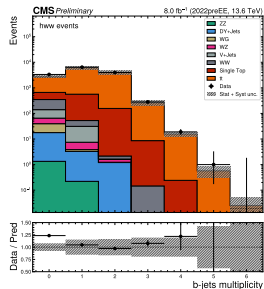


Top CR 2022postEE - extra MVA variables

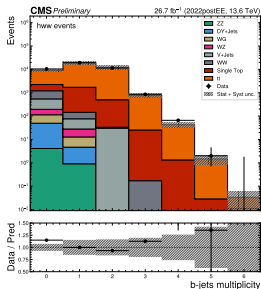


High Mass Control Region: `bjet_multiplicity`

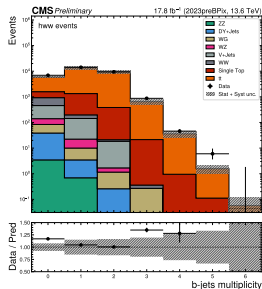
2022preEE



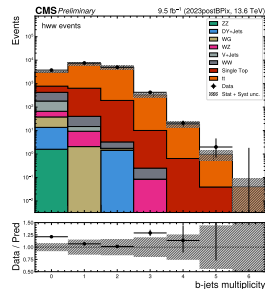
2022postEE



2023preBPix

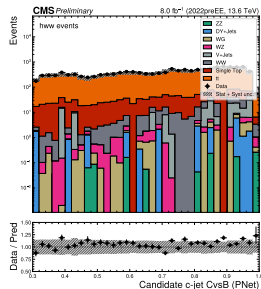


2023postBPix

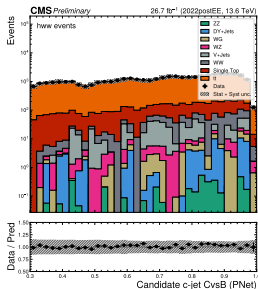


High Mass Control Region: cjet_cand_cvsvb_pnet

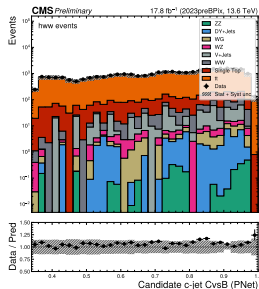
2022preEE



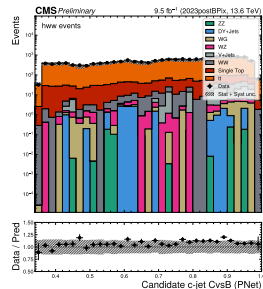
2022postEE



2023preBPix

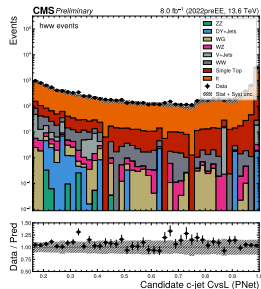


2023postBPix

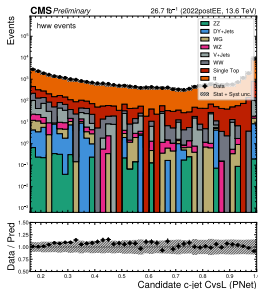


High Mass Control Region: cjet_cand_cvsl_pnet

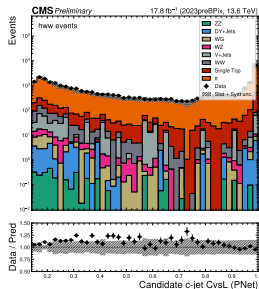
2022preEE



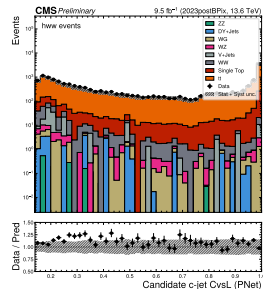
2022postEE



2023preBPix

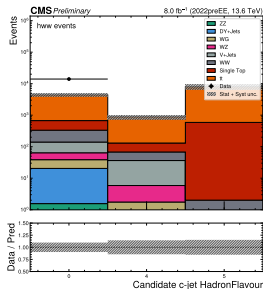


2023postBPix

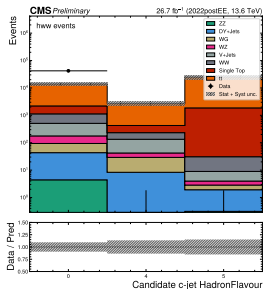


High Mass Control Region: cjet_cand_flavour

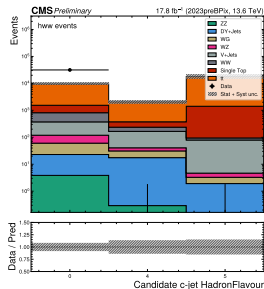
2022preEE



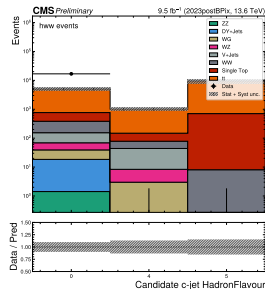
2022postEE



2023preBPix

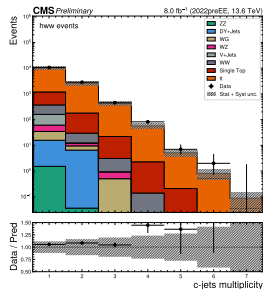


2023postBPix

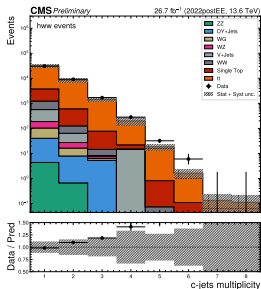


High Mass Control Region: cjet_multiplicity

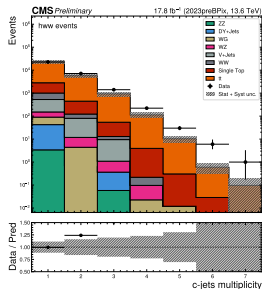
2022preEE



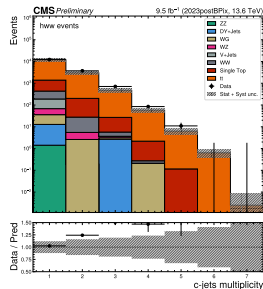
2022postEE



2023preBPix

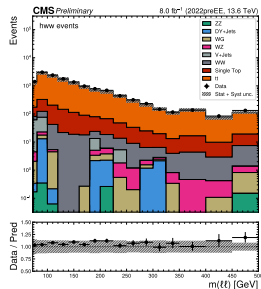


2023postBPix

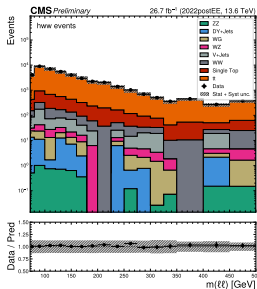


High Mass Control Region: dilepton_{mass}

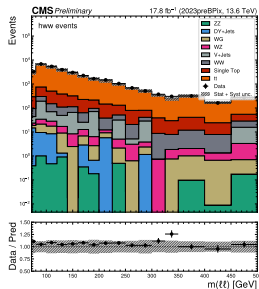
2022preEE



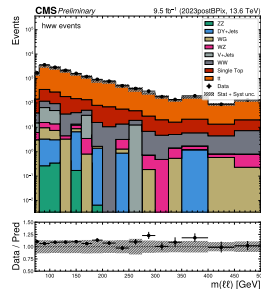
2022postEE



2023preBPix

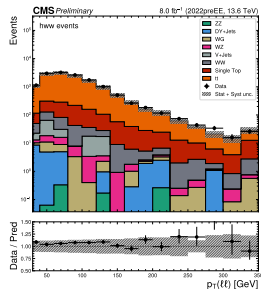


2023postBPix

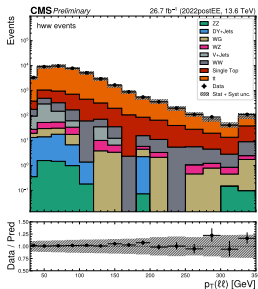


High Mass Control Region: dilepton_pt

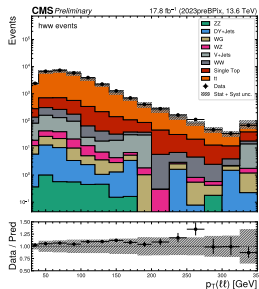
2022preEE



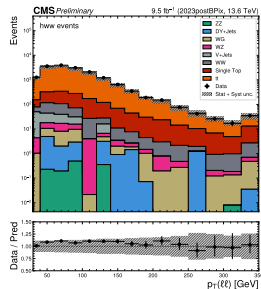
2022postEE



2023preBPix

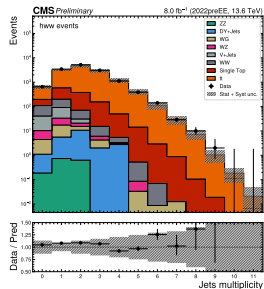


2023postBPix

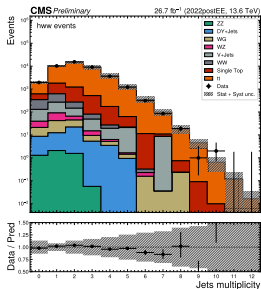


High Mass Control Region: jet_multiplicity

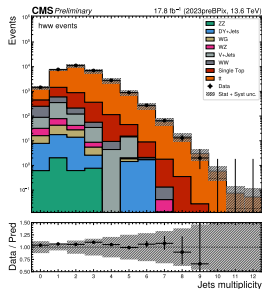
2022preEE



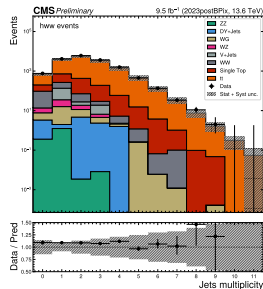
2022postEE



2023preBPix

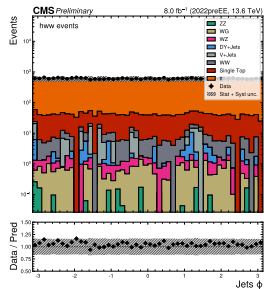


2023postBPix

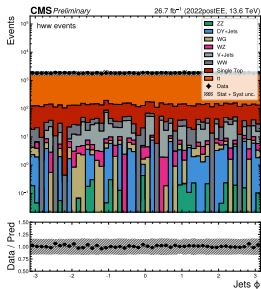


High Mass Control Region: jet_phi

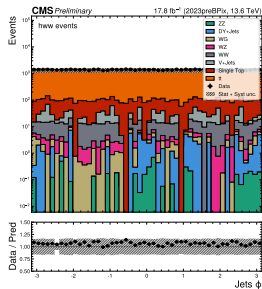
2022preEE



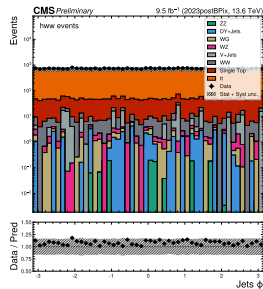
2022postEE



2023preBPix

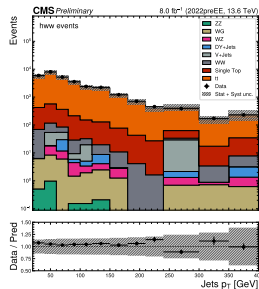


2023postBPix

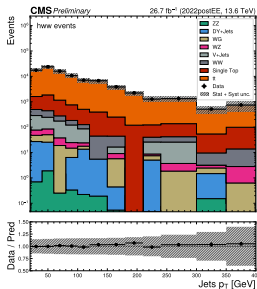


High Mass Control Region: jet_pt

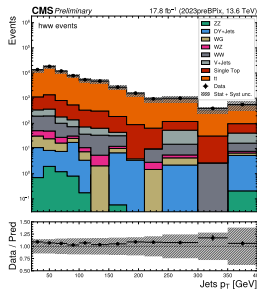
2022preEE



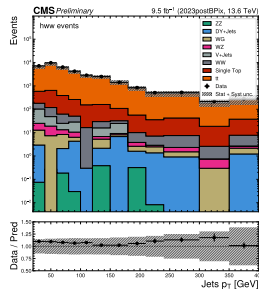
2022postEE



2023preBPix

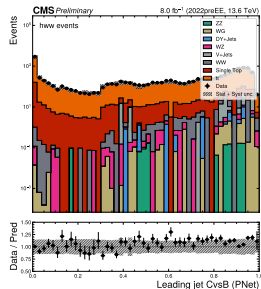


2023postBPix

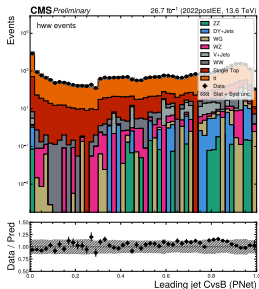


High Mass Control Region: leadingjet_cvsb_pnet

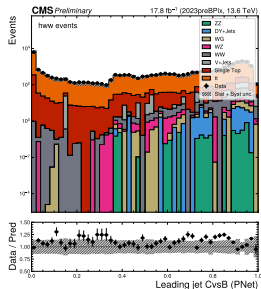
2022preEE



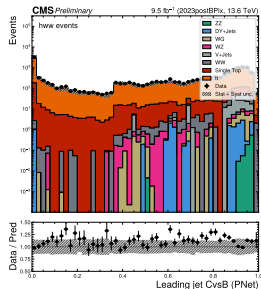
2022postEE



2023preBPix

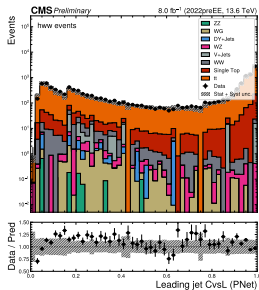


2023postBPix

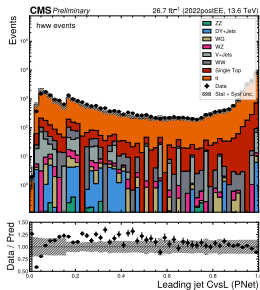


High Mass Control Region: leadingjet_cvsl_pnet

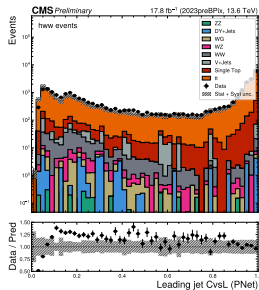
2022preEE



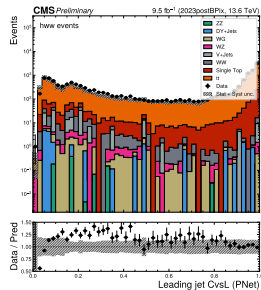
2022postEE



2023preBPix

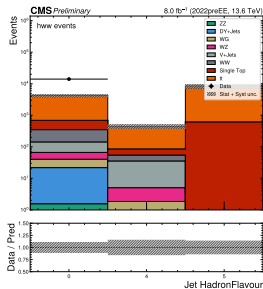


2023postBPix

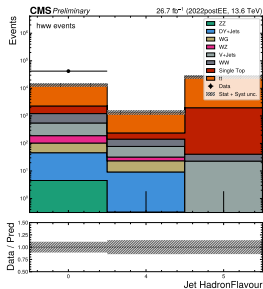


High Mass Control Region: leadingjet_flavour

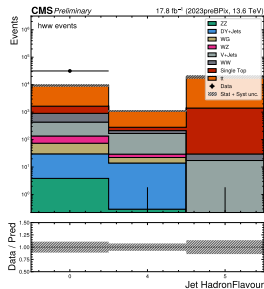
2022preEE



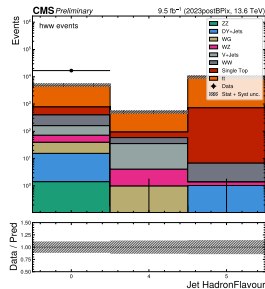
2022postEE



2023preBPix

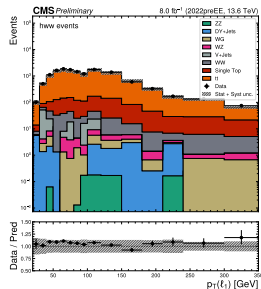


2023postBPix

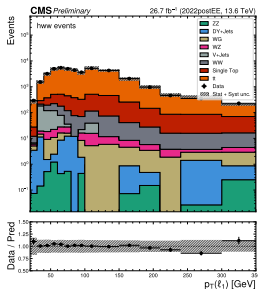


High Mass Control Region: lepton1_pt

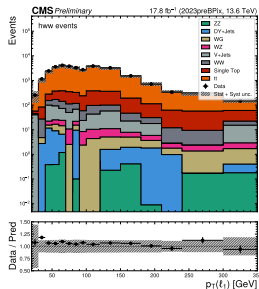
2022preEE



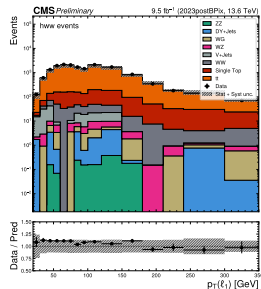
2022postEE



2023preBPix

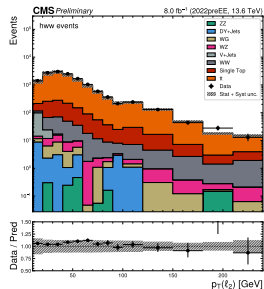


2023postBPix

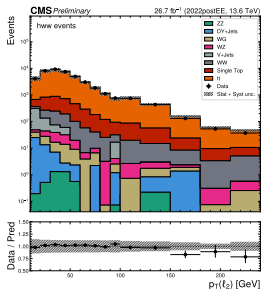


High Mass Control Region: lepton2_pt

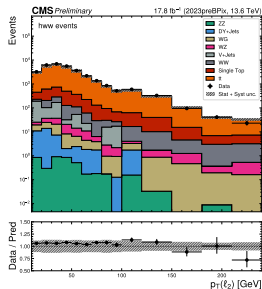
2022preEE



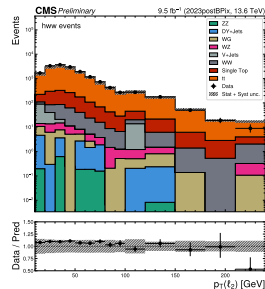
2022postEE



2023preBPix

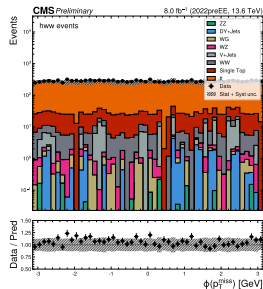


2023postBPix

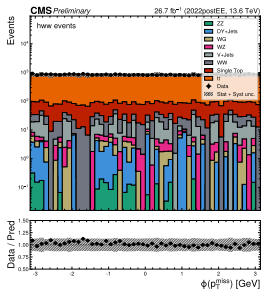


High Mass Control Region: met_phi

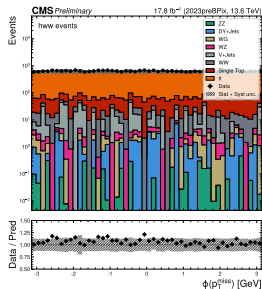
2022preEE



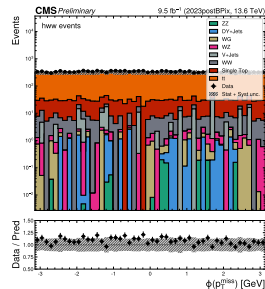
2022postEE



2023preBPix

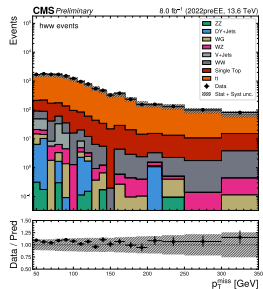


2023postBPix

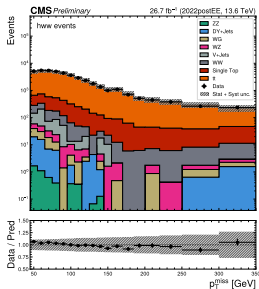


High Mass Control Region: met_pt

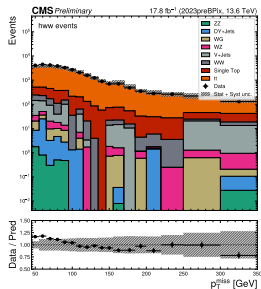
2022preEE



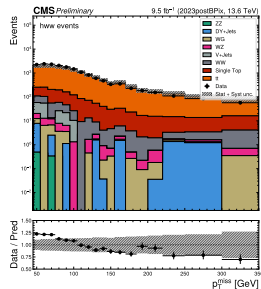
2022postEE



2023preBPix

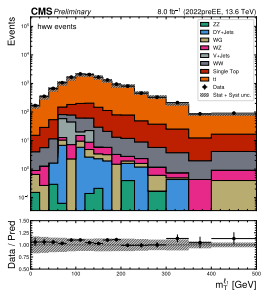


2023postBPix

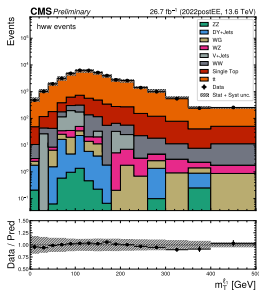


High Mass Control Region: mt11

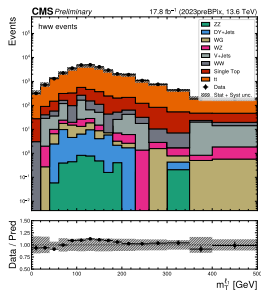
2022preEE



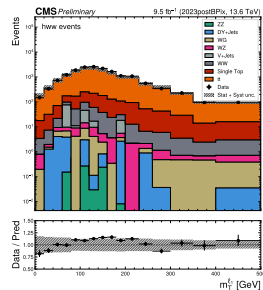
2022postEE



2023preBPix

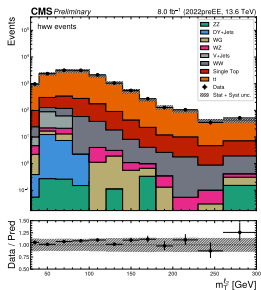


2023postBPix

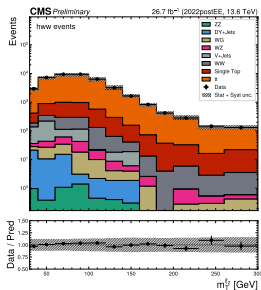


High Mass Control Region: $m_{\ell\ell}$

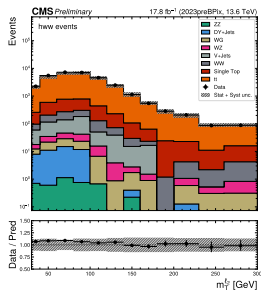
2022preEE



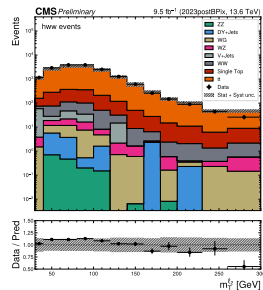
2022postEE



2023preBPix

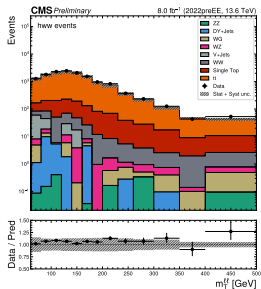


2023postBPix

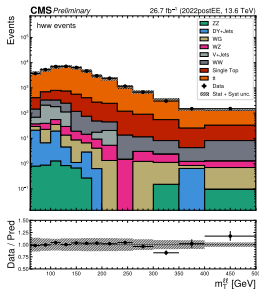


High Mass Control Region: mt11

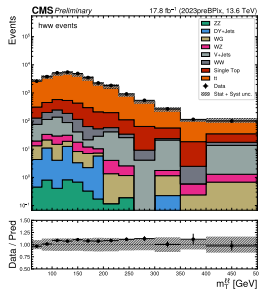
2022preEE



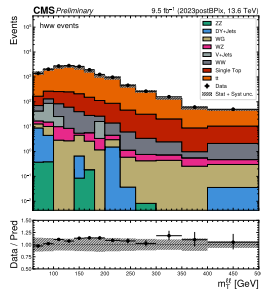
2022postEE



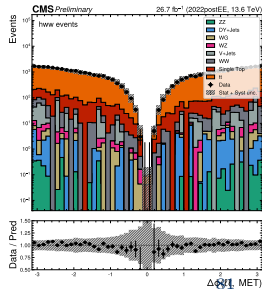
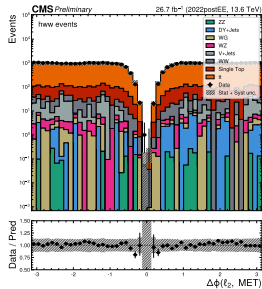
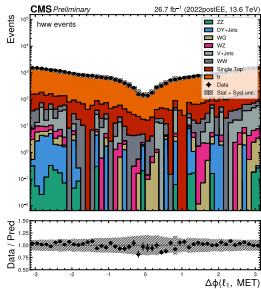
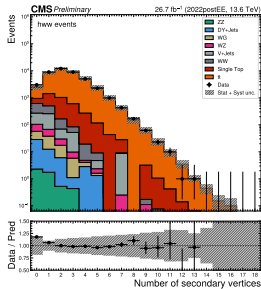
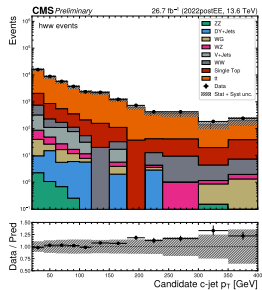
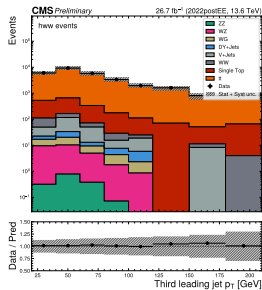
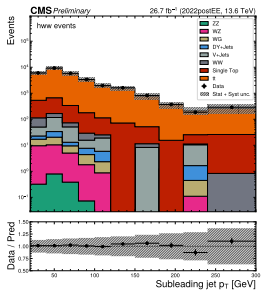
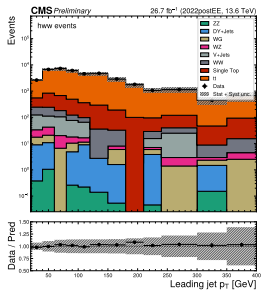
2023preBPix



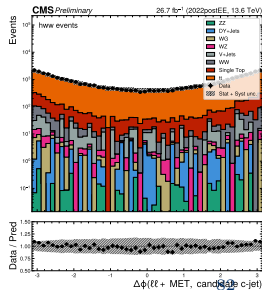
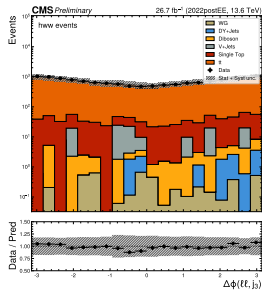
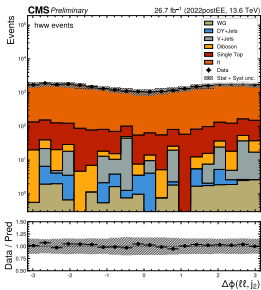
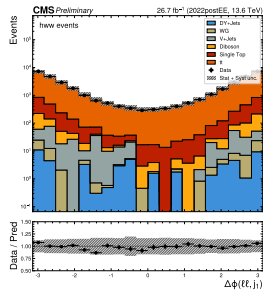
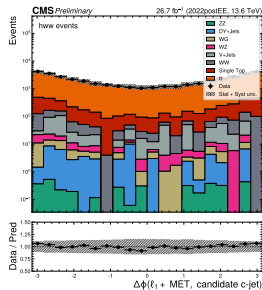
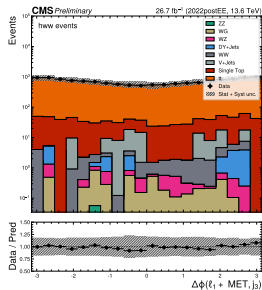
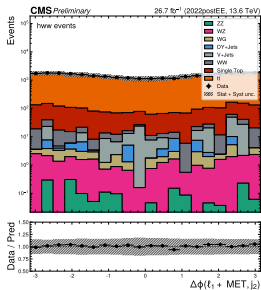
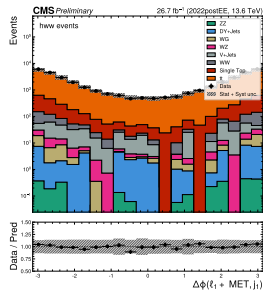
2023postBPix



High M CR 2022postEE - extra MVA variables



High M CR 2022postEE - extra MVA variables



High M CR 2022postEE - extra MVA variables

